AWS 5-Step Cleanup Process

Goal: Prevent **unwanted billing** and build **good AWS hygiene habits**.

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1 Delete Unused Security Groups

- Go to **EC2 > Security Groups**
- Check for any **custom groups** not attached to running instances.
- Select and **delete**.
  These are often left behind and can clutter your network setup.

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2 Release Elastic IPs

- Go to **EC2 > Elastic IPs**
- Identify **unassociated** IPs (not attached to instances).
- Select → **Actions > Release Elastic IP address**
  AWS charges for EIPs not in use!

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3 Remove Unused Key Pairs

- Go to **EC2 > Key Pairs**
- Identify **unused** SSH key pairs.
- Delete the ones you don't need anymore.
  Keep only those needed for future access.

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4 Delete Detached EBS Volumes

- Go to **EC2 > Volumes**
- Filter by **state = available**
- Select and **delete** all **detached** (unused) EBS volumes.
  Detached volumes still incur charges.

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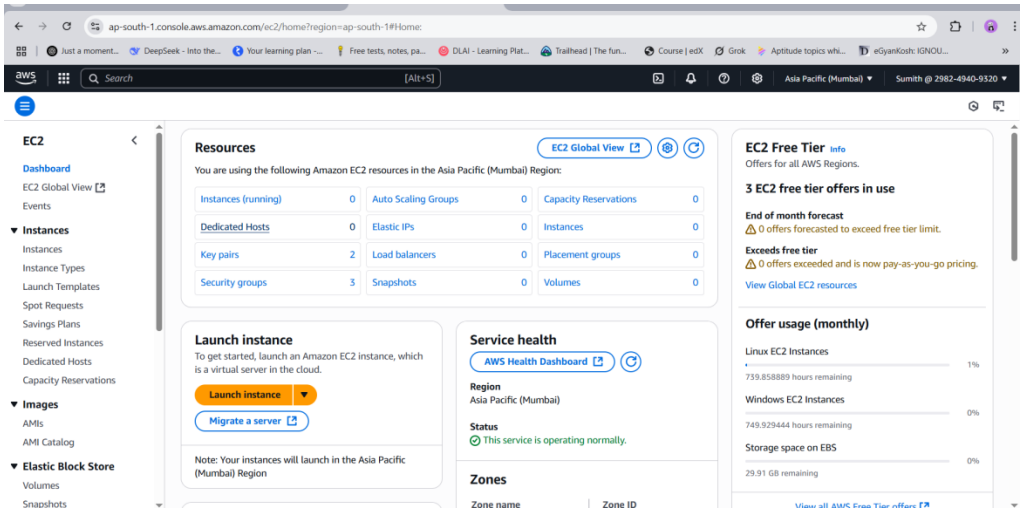
5 Verify Resource Dashboard

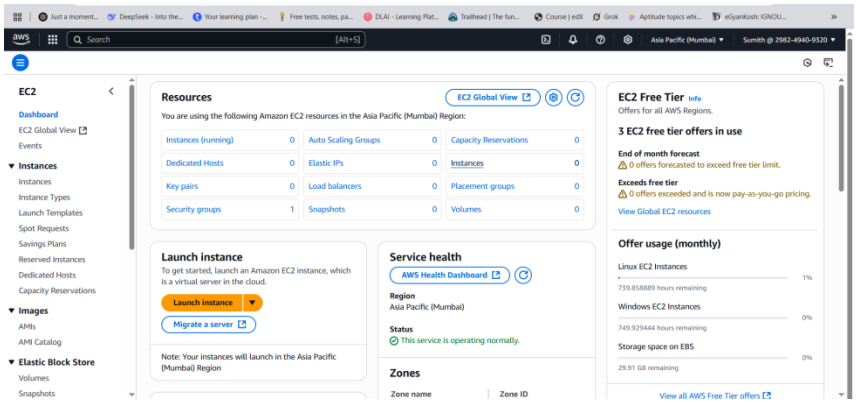
- Go to **EC2 Dashboard**
- Confirm:
  - ✅**Instances = 0**
  - ✅**Volumes = 0 available**
  - ✅**EIPs = 0 unassociated**
  - ✅**Snapshots, AMIs = cleaned (if not needed)**

  This final step ensures a **zero-footprint account**.

Dashboard before deletion.



Dashboard after deletion



Windows EC2 with IIS (via RDP)

1. **Launch EC2 Instance**
   - AMI: `Microsoft Windows Server 2019 Base`
   - Instance type: `t2.micro` (Free Tier)
   - Key Pair: Create or use existing `.pem`
   - Open port **3389 (RDP)** and **80 (HTTP)** in **Security Group**
2. **Connect via RDP**
   - Go to EC2 Console → Select instance → Click **Connect**
   - Download Remote Desktop file and get password using key pair
   - Use **RDP** to log in
3. **Install IIS (Internet Information Services)**
   - Open **Server Manager** → Add Roles and Features
   - Select **Web Server (IIS)** → Proceed with default settings and install
4. **Test IIS**
   - Open browser on your system
   - Visit: `http://<Public IP>` → You should see the **IIS welcome page**

Linux EC2 with Apache (via SSH)

1.
2. **Launch EC2 Instance**
   - AMI: `Amazon Linux 2023` or `Ubuntu`
   - Instance type: `t2.micro`
   - Key Pair: Use `.pem` file
   - Open port **22 (SSH)** and **80 (HTTP)** in **Security Group**
3. **Connect via SSH**
   - Use Terminal or Git Bash:

```
bash
CopyEdit
ssh -i your-key.pem ec2-user@<Public-IP>   # Amazon Linux
# OR
ssh -i your-key.pem ubuntu@<Public-IP>     # Ubuntu
```

4. **Install Apache**
   - For Amazon Linux:
     ```
     bash
     CopyEdit
     sudo yum update -y
     sudo yum install httpd -y
     sudo systemctl start httpd
     sudo systemctl enable httpd
     ```
   - For Ubuntu:
     ```
     bash
     CopyEdit
     sudo apt update
     sudo apt install apache2 -y
     sudo systemctl start apache2
     sudo systemctl enable apache2
     ```
5. **Test Apache**
   - Visit: `http://<Public IP>` → You should see the **Apache default page**

1. AWS Free Tier Limits — What & How Much?

   Types of Free Tier Offers:

Type	Duration	Examples
Always Free	Unlimited	Lambda, DynamoDB, S3 Glacier, CloudWatch
12-Month Free	12 months	EC2 t2.micro/t3.micro, S3, RDS, CloudFront
Trials	Limited time	SageMaker Studio Lab, AWS Support plans, etc.

   Key Services and Usage Thresholds:

Service	Free Tier Limit (per month)	Duration
EC2	750 hrs/month of **t2.micro / t3.micro**	12 months
S3	5 GB Standard Storage	12 months
RDS	750 hrs + 20 GB SSD + backups	12 months
Lambda	1 million requests + 400,000 GB-seconds	**Always Free**
CloudFront	1 TB of data transfer	12 months
DynamoDB	25 GB storage + 200M requests	**Always Free**
CloudWatch	10 custom metrics + 5GB logs	**Always Free**

2. Deep Dive into Protocols & Ports:

Protocol	Port	Use Case	OS
HTTP	80	Web server (unsecured)	Windows/Linux
HTTPS	443	Web server (secured)	Windows/Linux
SSH	22	Remote login (CLI)	Linux
RDP	3389	Remote desktop access	Windows

Experiment: Security Group Modification

✅ ***Allowing HTTP, HTTPS***

1. Go to **EC2 > Security Groups**
2. Edit **Inbound Rules**:
    - Allow **Port 80 (HTTP)** – Anywhere (0.0.0.0/0)
    - Allow **Port 443 (HTTPS)** – Anywhere

✅ ***Allow RDP (Windows)***

- Add rule: **Port 3389**, Protocol **TCP**, Source **My IP**

✅ ***Allow SSH (Linux)***

- Add rule: **Port 22**, Protocol **TCP**, Source **My IP**

❌ ***Block Access (for hardening)***

- Remove or restrict rules to IPs you trust